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BeyondDeskVR: an extended virtual hand interaction system in virtual reality

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ABSTRACT
As virtual reality (VR) technology advances, the use of VR scenarios and applications is becoming increasingly prevalent, and the fatigue issues associated with existing mid-air hand interactions are becoming more pronounced. To alleviate fatigue in VR interactions, we developed a desktop interaction system, BeyondDeskVR. Designed for seamless integration with mid-air hand interaction in seated VR work environments, the system enables portable interaction input and mode switching. By extending virtual hand interactions to the desktop space, the system could be used on any desktop. We utilised infrared laser projection for desktop gesture recognition and achieved seamless transitions between desktop gesture interactions and mid-air hand interactions in typical three-dimensional scenarios, including pointing and selection, user interface interaction, and object manipulation. We conducted a comprehensive evaluation of the BeyondDeskVR gesture interaction system and mid-air hand interaction using a pointing selection task, assessing their performance in terms of subjectivity, performance, and fatigue. The results indicate that the BeyondDeskVR system has high user acceptance, significantly lower fatigue than does mid-air hand interaction, and comparable interaction efficiency and accuracy. Integrating this system with mid-air hand interaction can effectively reduce fatigue in VR interactions and enhance the overall VR experience.

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Gesture interaction system; finger touch; laser projection; virtual reality; interaction device

1. Introduction

As virtual reality (VR) technology advances, the use of VR scenarios and applications is becoming increasingly widespread. Hands are the most important effectors for humans, and various mid-air hand interaction methods, including virtual hand touch, mid-air gestures, and ray casting, have become mainstream interaction techniques in VR. At present, mid-air hand interaction has been widely applied across diverse use cases, such as gaming and entertainment (Moon, Orr, and Jeon 2023), surgery (McCloy and Stone 2001), classroom teaching (Soliman et al. 2021), computer-aided design (Wang et al. 2021), and mobile offices (Aufegger, Elliott-Deflo, and Nichols 2022; Ofek et al. 2020).

For enhanced freedom in three-dimensional control and display space, mid-air hand interaction offers advantages such as being intuitive, flexible, and closely aligned with natural human behavioural patterns (Koutsabasis and Vogiatzidakis 2019). With recent VR devices such as the Oculus Quest 2, which integrates cameras into a headset, users can achieve convenient mid-air hand interactions without additional equipment. However, mid-air hand interaction has potential drawbacks, such as arm fatigue (Iqbal et al. 2021), insufficient accuracy (Jones, McIntyre, and Harris 2020), and a lack of controllability (Mendes et al. 2019). These issues are reflected in the fact that prolonged arm lifting by users causes muscle contraction, leading to arterial constriction and impeding blood flow. Consequently, users experience sensations of fatigue and muscle soreness, reducing their ability to accurately select targets (Reynaert et al. 2023); this phenomenon is known as the 'gorilla arm effect' (Li et al. 2019).

To address the aforementioned issues, several researchers have introduced the concept of contact-based gesture interaction. This approach relies on wearable devices, force or tactile sensors, electromyography (EMG) sensors, touchscreens, and other technologies to improve the accuracy and efficiency of gesture interactions. Several contact-based interaction methods involve physical contact with surfaces, which effectively reduce arm fatigue. However, challenges such as difficult localisation and cumbersome configurations often occur in such interactions, making it difficult to implement immersive VR. Overall, the arm fatigue issue in mid-air hand interactions, as well as the intrusiveness and localisation difficulties associated with contact-based interactions, make it challenging to sustain

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bare-hand interactions in VR in a long-term and stable manner.

The seated desktop scenario is considered to be one of the most common application contexts in human-computer interactions (HCIs) (Lu et al. 2018; Mcgill et al. 2020). With the introduction of concepts such as portable virtual offices (Ruvimova et al. 2020) and fully immersive desktops (Zielasko et al. 2017), it has become important for individuals to wear VR headsets to carry out low-fatigue, high-frequency interactions in a seated state. The desktop environment naturally provides a physical surface for desktop gesture interaction, which makes it possible to integrate the advantages of desktop gestures and mid-air hand interaction, and realise broader applications in more extensive spatial contexts.

In summary, we propose BeyondDeskVR, an extended virtual hand interaction system that integrates contact-based and noncontact-based interactions, to address the issue of fatigue in mid-air hand interactions within seated desktop environments. We implemented mid-air hand interaction via sensors from a commercial head-mounted display. Building upon this, our system introduces desktop gesture interactions via an independently designed sensing device. It allows for interaction on any desktop surface via minimal finger movement, relying on 3D spatial cursor feedback without complex configuration steps to complete interactions. The system seamlessly alternated between desktop gesture interactions and mid-air hand interactions in real time, alleviating user fatigue during use and fulfilling the comprehensive requirements of desktop scenarios, including spatial freedom, speed, precision, and fatigue reduction in the interaction space.

The remainder of this paper is structured as follows. Section 2 offers a literature review; Section 3 outlines the principles of the BeyondDeskVR system; Section 4 conducts an ergonomic evaluation experiment and describes the results; Section 5 provides a discussion; Section 6 outlines the limitations and future work; and Section 7 provides a concluding summary.

2. Related works

A usable HCI system requires reliable input devices, well-designed interaction methods, and effective evaluation and validation procedures to establish its usability. In this section, we expound on three aspects: the input devices, interaction methods, and evaluation methods.

2.1. Input devices in VR

Input devices are integral components of virtual environment interaction systems. Serving as physical carriers for facilitating various interaction modalities, they have a direct effect on the communication constraints between users and virtual environments (VEs) (LaViola et al. 2017). From the system control perspective, selecting an appropriate input device for specific task contexts is crucial. Although 2D input devices for GUIs, such as keyboard and mouse devices (Fedor et al. 2021) and touch-based devices (Srinivasan et al. 2020), are capable of achieving continuous, stable, and precise two-dimensional input, their usability decreases when they are utilised in VEs through blind operation methods (Knierim et al. 2018).

Currently, the mainstream input devices utilised in VEs primarily consist of 3D controller-type devices (De Paolis and De Luca 2020) and vision-based sensing devices (Jones, McIntyre, and Harris 2020; Li et al. 2019). 3D controller-type devices allow real-time spatial tracking at high refresh rates (Li et al. 2022) and are often enhanced with additional physical components (e.g. buttons, joysticks, and touchpads) to improve functionality. Vision-based sensing devices typically employ devices such as RGB or infrared cameras to track localised or full-body user gestures and movements to achieve three-dimensional control (Mahdikhanlou and Ebrahimnezhad 2023). In summary, these devices rely primarily on three-dimensional limb movements in mid-air to meet the high-DOF spatial interaction requirements (Huang et al. 2021); however, suboptimal interaction postures with the arms raised are more likely to induce operator fatigue (Jang et al. 2017), accompanied by issues such as shaking and reduced precision.

In recent years, researchers have proposed several novel contact-based virtual hand input devices for desktop scenarios, most of which are anchored to desktop or tangible physical surfaces to facilitate interaction. These devices can be classified on the basis of their foundational principles, including mobile smart devices, physical keyboards, force feedback tactile sensing devices, and desktop-based vision sensing devices. Among these, the application of mobile smart devices for tangible interface interactions has emerged as the main research focus in this field. Many researchers have adapted mobile smart devices, including tablets (Biener et al. 2020; Le et al. 2021; Surale et al. 2019), smartphones (Shin and Lee 2022; Zhang et al. 2023), and smartwatches (Zhang et al. 2022), to accurately record user finger information or touch gestures, which are subsequently mapped to remote pointing or system control commands. Several researchers have introduced physical keyboards into desktop contexts to facilitate text input activities (Knierim et al. 2018; Menzner et al. 2019). Force feedback tactile sensing devices are common active sensors (LaViola et al. 2017) that utilise inertial measurement unit (IMU) sensors to capture data on the user's finger or wrist joint motion, enabling the recognition of user actions on the desktop. These devices, such as TapID devices (Meier et al. 2021), are highly accurate but require wearability and possess a certain level of invasiveness. Desktop-based vision-sensing devices, such as Vrrrroom (Sousa et al. 2017), MRTouch (Xiao et al. 2018), and ShawdowTouch (Liang et al. 2023), utilise cameras to record user finger movements within the desktop area or to extract touch interactions on the surface. In summary, these desktop-based input devices rely on physical surfaces for support to avoid operation fatigue to a certain extent, suggesting the potential for prolonged use.

Additionally, some studies have employed multimodal approaches to achieve natural input in virtual reality environments. These studies have integrated external sensors such as eye trackers, motion capture systems, electromyography sensors, brain-computer interfaces, and voice sensors to construct multimodal interaction systems (Wen et al. 2020), or have combined 3D controllers with barehand virtual hand interactions to enable efficient input (Capece et al. 2021). In the future, the development of multimodal interaction systems (Cao et al. 2023) will be a critical research direction for achieving efficient and natural inputs in virtual reality.

2.2. Interaction methods in VR

Interaction methods refer to the mapping between the information gathered by input devices and the system control mechanisms within VEs. Typically, one input device can support multiple interaction methods. Establishing a natural, efficient, and suitable mapping relationship between input devices and the display space is often essential for user-friendly interactions (LaViola et al. 2017). From the perspectives of display and control, a GUI typically uses control-display mapping, which is two-dimensional to two-dimensional (Mendes et al. 2019). However, as VEs expand the display area into three-dimensional space, interaction methods undergo significant transformations.

In 3D VEs, users have the option to use either direct metaphors (Figueiredo et al. 2018) or indirect metaphors (Brasier et al. 2020) to accomplish most interaction tasks, which fall into categories such as pointing selection (Kim, Park, and Kim 2020), manipulation (Goh, Sunar, and Ismail 2019), navigation (Carbonell-Carrera, Saorin, and Jaeger 2021), and system control (Reski and Alissandrakis 2020). With respect to direct metaphors, mid-air hand interactions, such as the typical virtual hand metaphor (McAnally, Wallwork, and Wallis 2023) and the GO-GO metaphor (Poupyrev et al. 1996), are typical direct interaction methods. Direct metaphors require users to interact in a self-centred way and are characterised by an intuitive nature, a strong sense of immersion, and immediate feedback. However, since mid-air hand interactions depend on the spatial movements of the limb raised in unsupported spatial conditions, limb fatigue (Iqbal et al. 2021) and occlusion issues (Lu et al. 2018) between sensors and limb movements inevitably arise. Several studies have attempted to reduce operational fatigue by relocating interaction targets to desktops and positioning visual interaction elements on the table (Zielasko et al. 2019); however, this approach can cause neck fatigue. In addition to virtual hand metaphors, direct metaphors include ray metaphors, such as ray casting (Wagner, Stuerzlinger, and Nedel 2021), flashlights (Liang and Green 1994), and fishing rods (Funk et al. 2019).

With respect to indirect metaphors, to reduce fatigue and enhance interaction efficiency in VEs, several studies have proposed indirect metaphor interaction methods (Findlater et al. 2017; Forlines et al. 2007; Iqbal et al. 2021), which separate the input and display areas to enable large-scale visual object interactions within a more comfortable input space. Some indirect metaphorical techniques, such as Virtual Pad (Andujar and Argelaguet 2007), facilitate the indirect control of targets by relocating motion space to lower fatigue interaction areas (Brasier et al. 2020; Kim and Xiong 2022). Additionally, other methods utilise third-person perspective control of virtual environment agents, focusing on interactions within limited spaces, such as Worlds in Miniature (WIM) (Stoakley, Conway, and Pausch 1995) and Voodoo Doll (Pierce, Stearns, and Pausch 1999). Moreover, previous studies have examined techniques such as implementing miniature touch-sensitive regions (Oh, Park, and Park 2020) and developing microgestures (Li et al. 2021) to restrict a user's operational range, ultimately enhancing interaction effectiveness. While indirect metaphors can effectively mitigate fatigue issues associated with prolonged mid-air hand interactions, they may also impact user comfort (Lou et al. 2021). Therefore, careful consideration of specific task contexts is necessary when designing suitable interaction methods for applications.

2.3. Evaluation methods in VR

System evaluation is conducted to assess whether the system satisfies design objectives, identify potential shortcomings, and provide suggestions for system enhancement (Chen and Wu 2023). The evaluation of interaction systems in VEs generally incorporates metrics such as subjective assessments (Kim, Park, and Kim 2020), task performance (Brasier et al. 2020) and interaction fatigue (Satriadi et al. 2019).

In terms of subjective assessments, user experience and comfort are often quantified through subjective scales such as the NASA-TLX (Jones, McIntyre, and Harris 2020), SUS (Jones, McIntyre, and Harris 2020), and Borg scales (Satriadi et al. 2019). In terms of task performance, the user experience and user comfort can also be quantified by efficiency and effectiveness. Specifically, interaction accuracy and speed are two common metrics used to evaluate user performance. Interaction accuracy measures user performance by recording error rates (Li, Cho, and Wartell 2018) or offset levels (Iqbal et al. 2021; Meier et al. 2021) during tasks, whereas interaction speed is typically quantified by recording the time taken to complete tasks (Bakar et al. 2023; Hansberger et al. 2017). For general performance evaluation, Fitts's law (Fitts 1954) serves as a means to assess the overall accuracy and speed of user interaction via metrics such as throughput (TP) (MacKenzie 1992). The Fitts Law paradigm, which is based on ISO 9241-411 (Iso 2012), has also been utilised by researchers to evaluate pointing selection techniques (Forlines et al. 2007; Kim, Park, and Kim 2020; Kim and Xiong 2022; McAnally, Wallwork, and Wallis 2023). In terms of interaction fatigue, researchers have recently started incorporating physiological indicators or utilising computer vision techniques to evaluate user fatigue. Several researchers have analysed muscle fatigue via invasive electromyography (EMG) sensors (Penumudi et al. 2020). Others have used noninvasive visual sensing devices to develop arm fatigue assessment software (Hincapié-Ramos, Guo, and Irani 2014; Jang et al. 2017), which directly quantifies fatigue levels on the basis of human kinetics (Sidenmark and Gellersen 2020; Wang et al. 2018) or the consumed endurance (CE) derived from the human biomechanical structure (Hincapié-Ramos et al. 2014).

2.4. Summary

In summary, to achieve low-fatigue, high-frequency, and long-term seated interactions with extended virtual hand interactions, it is essential to provide arm support on the desktop. The design of interaction methods can draw inspiration from indirect metaphors, transferred input areas, and narrowed interaction ranges. A comprehensive performance evaluation should encompass multiple dimensions of subjectivity, performance, and fatigue.

3. BeyondDeskVR system

To achieve prolonged virtual hand interaction in VR, we proposed a desktop gesture interaction system. On the basis of mid-air hand interactions, this system leveraged desktop space to accomplish contact-based gesture interactions. Upon the integration of this system, VR devices are capable of supporting the manipulation of 3D visual objects under both spatial 3D and desktop 2D inputs. On the basis of the desktop context within the system, we referred to this interaction system as BeyondDeskVR. As shown in Figure 1, users wear a VR headset and sit at any desk with this system. Users can freely switch between mid-air hand interaction and desktop gesture interaction states. Specifically, desktop gesture interaction is enabled by the BeyondDeskVR system's desktop gesture recognition device, whereas mid-air gesture recognition is accomplished using the front-facing cameras of the head-mounted display. In the desktop interaction state, users control the movements of the virtual cursor by moving their fingers on the desktop indirectly, enabling them to point to interactive targets in the virtual interface. Target selection, user interface (UI) interactions and object manipulation tasks are accomplished through multiple touch gestures. The system was created through a six-step process that included: constructing the needed desktop gesture recognition hardware, detecting the touch states of fingers, acquiring the world coordinates of the touching fingers, mapping control and display, defining pointing selection strategies, and establishing an application prototype for extended virtual hand interactions that combines mid-air hands and desktop gestures. Detailed explanations of the six stages are provided in Sections 3.1–3.6.

3.1. Hardware construction

Hardware construction aims to provide the system with a device that can accurately recognise the user's hand gestures on the desktop, which forms the foundation for subsequent interactive functions. The hardware setup employed the principles of infrared laser projection sensing technology (Korth 1998), with an infrared laser projection module and an infrared image capture module. The infrared laser projection module was equipped with a linear 980 nm infrared laser transmitter placed flat against the base and desktop, which emitted a layer of parallel infrared light onto the surface of the desktop. The infrared image capture module was equipped with a 120° wide-angle infrared camera that could receive 980 nm infrared light, which was used for capturing the desktop scene image. When the

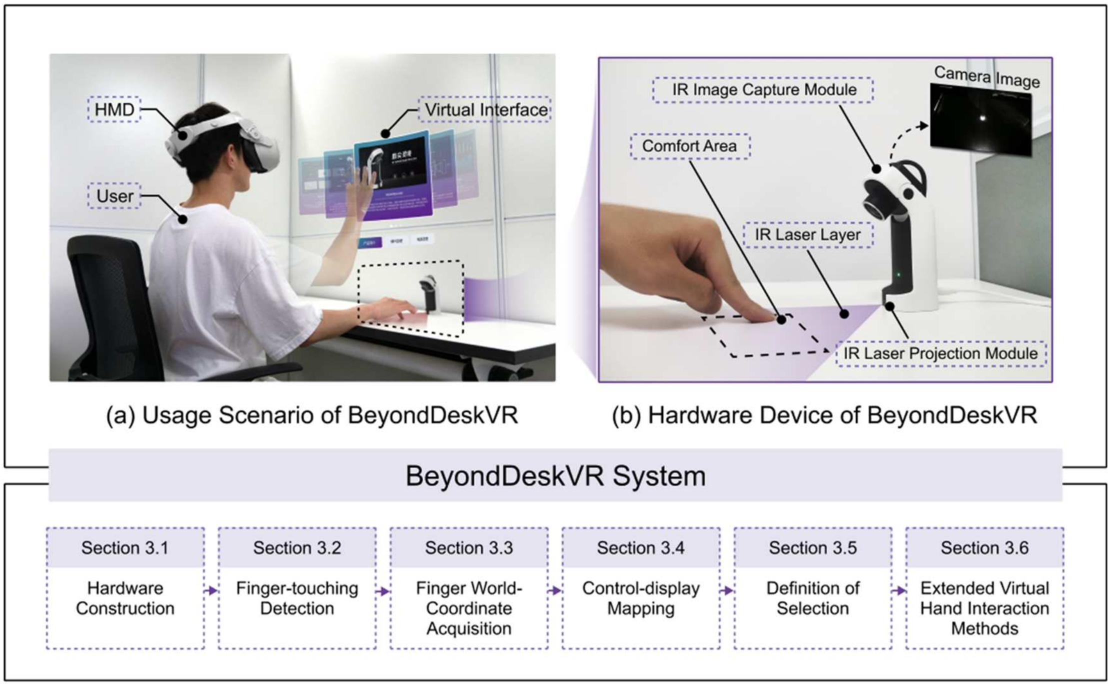


Figure 1. The technical implementation principle of BeyondDeskVR.

user's finger touched the desktop, the infrared light at the point of contact was intercepted and diffusely reflected off the finger to the infrared camera, creating a light spot in the captured image. By analysing and processing these light spots frame by frame, the system could achieve real-time identification of the user's touch gestures on a desktop.

To establish a suitable clicking area, we recorded the comfortable desktop interaction region for 10 participants. Through this process, we determined that the user's comfortable single-handed interaction area was concentrated within a rectangular area measuring approximately 34 cm in length and 17 cm in width. Additionally, an adjustment pivot was integrated into the image capture module, enabling the infrared camera to rotate within a range of 0–50 degrees for angle adjustments during shooting. This device also incorporated a microcontroller with power and communication modules, as well as indicator lights, and was designed with structural components and a casing. Finally, the hardware device was manufactured via CNC machining, resulting in final dimensions of 127 mm in height, 45 mm in length, and 47 mm in width.

3.2. Finger-touching detection

Finger-touching detection aims to accurately identify the touch states of fingers and extract their positions on the basis of desktop images taken by the infrared image capture module. Once the hardware captured the desktop images, they were transmitted to the computer for frame-by-frame analysis via a touch detection algorithm based on image processing. This process enables the detection of finger touch states within the images, as shown in Figure 2.

The finger-touching detection algorithm utilised the OpenCV library for image processing. During device initialisation, the first stable frame image was recorded as the background image. Afterwards, the algorithm calculated the difference between the current received

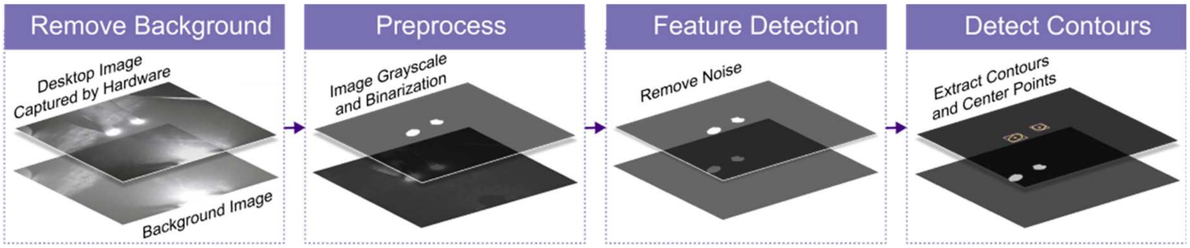


Figure 2. Finger-touching detection process.

desktop image and the background image, eliminating static interference such as noise from complex desk environments. The background-removed image was converted to grayscale and then binarized for touch point feature extraction. However, the processed image still contained dynamic noise generated randomly by finger movements during user interactions. The algorithm utilised a thresholding method to eliminate image noise that did not meet the conditions and retained the light spot that corresponds to finger touching. Following the removal of image noise, the algorithm extracted the contour of the user's finger-touching point and recorded the pixel coordinates of the centre of the contour's bounding rectangle in the image as the location of the finger touch point. This concluded the user's finger-touching detection process.

3.3. Finger world-coordinate acquisition

Finger world-coordinate acquisition aims to transform the finger centre's pixel coordinates obtained through image processing into the user's actual finger position on the desktop, which provides a foundation for subsequent mapping control and display. The infrared camera's tilted placement and 120° shooting angle caused image distortion. Specifically, close-up areas appeared larger with minimal distortion, whereas distant areas appeared smaller with significant distortion, creating the visual effect of 'objects appearing larger when closer to the camera'. However, the finger-touching point positions identified after processing the image were pixel coordinates within the image, thereby increasing susceptibility to image distortion and preventing us from precisely reflecting the finger positions in the real world. To ensure an accurate mapping relationship between the finger's movement space in the real world and the cursor's display space in the virtual environment, it was necessary to acquire the exact positions of the fingers in the physical realm and overcome the issue of image distortion.

During finger position capture via hardware, the user's finger-touching point underwent transformations across four coordinate systems: the world coordinate system $O_W - X_W Y_W Z_W$, the camera coordinate system $O_C - X_C Y_C Z_C$, the image coordinate system $O_1 - xy$, and the pixel coordinate system $O_0 - uv$. These transformations ultimately determine the pixel coordinates within the image, as shown in Figure 3.

To transform the pixel coordinates obtained through image processing into more accurate and easily interpretable world coordinates, it is essential to establish the transformation relationship between the world coordinates and the pixel coordinates of the finger-touching point. The coordinate transformation process from the world position (X_W, Y_W, Z_W) of the finger-touching point to the pixel coordinate (u, v) is as follows:

3.3.1. Transformation from world coordinates to camera coordinates

The world coordinates (X_W, Y_W, Z_W) of the finger-touching point, after undergoing translation and rotation with respect to the coordinate axes, are transformed into camera coordinates (X_C, Y_C, Z_C) relative to the camera. The transformation relationship is expressed in Equation (1). Here, R represents the rotation matrix during the coordinate transformation process, and T represents the translation matrix during the coordinate transformation process. Since the user's finger remains in a single plane while touching the desktop, set $Z_W = 0$.

$$\begin{bmatrix} X_C \\ Y_C \\ Z_C \end{bmatrix} = R \begin{bmatrix} X_W \\ Y_W \\ 0 \end{bmatrix} + T \tag{1}$$

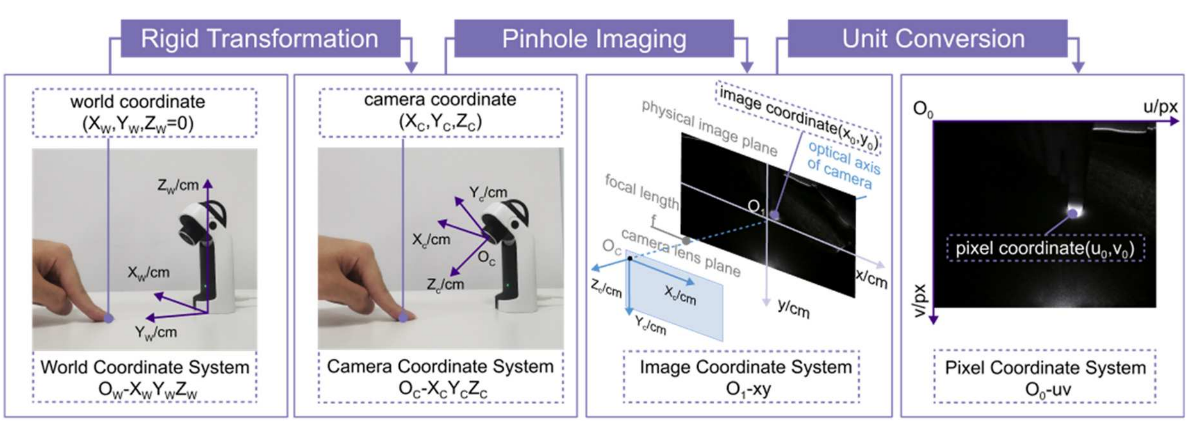


Figure 3. The process of coordinate transformation of finger imaging.

3.3.2. Transformation from camera coordinates to image coordinates

wThecameracoordinates(X_C, Y_C, Z_C)ofthefinger-touchingpointareprojectedontothecamera's focalplaneviathe pinhole imaging principle (Dawson-Howe and Vernon 1994),andtheimagecoordinates(x, y)ofthefinger-touching point are obtained. The transformation relationship is expressed in Equation (2). Here, f represents the camera's focal length.

$$Z_C\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} X_C \\ Y_C \\ Z_C \\ 1 \end{bmatrix} \tag{2}$$

3.3.3. Transformation from image coordinates to pixel coordinates

The physical image on the camera's focal plane needs to be converted into a standard image using pixel units. At this point, the image coordinates (x, y) of the finger-touching point are ultimately transformed into pixel coordinates (u, v), which are the coordinates obtained during image processing, as described in Section 3.2. The transformation relationship is expressed in Equation (3). Here, k_x and k_y represent the scaling factors that convert physical units to pixel units, and (u_0, v_0) represents the pixel coordinates of the centre point of the user's finger.

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} k_x & 0 & u_0 \\ 0 & k_y & v_0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \tag{3}$$

In summary, the overall coordinate transformation relationship between the world position of the finger-touching point (X_W, Y_W, Z_W) and the pixel coordinate (u, v) is expressed in Equation (4). Here, $\begin{bmatrix} f_x & 0 & u_0 & 0 \\ 0 & f_y & v_0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$ represents the camera intrinsic matrix, which was determined in this paper via the MATLAB camera calibrator toolbox based on Zhang Zhengyou's checkerboard calibration method (Zhang 2000); $\begin{bmatrix} R & T \\ 0 & 1 \end{bmatrix}$ represents the camera extrinsic matrix, which was determined by calculating the translation and rotation relationships between the camera centre and the world coordinate system centre. Since the camera in the hardware setup remains relatively fixed with respect to the world coordinate system, this step needs to be repeated only once. The value of Z_C can be determined using Equation (1), and it is dependent only on X_W, Y_W, the rotation matrix R, and the translation matrix T.

$$Z_C\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & 0 & u_0 & 0 \\ 0 & f_y & v_0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} R & T \\ 0 & 1 \end{bmatrix}\begin{bmatrix} X_W \\ Y_W \\ 0 \\ 1 \end{bmatrix} \tag{4}$$

On the basis of the principles described above, we could reverse-engineer the user's actual finger-touch position based on the pixel coordinates of the finger highlight obtained through image processing. In situations where the user's actual finger position (X_W, Y_W, 0) is known, we can establish a proportional control-display mapping within the real-world motion space.

3.4. Control-Display mapping

Control-display mapping aims to establish the mapping relationship between the world coordinate of the finger centre on the desktop and the pointing direction vector in virtual space, thereby determining the target that the user intends to select. To enable system control from a 2D desktop motion space supporting 3 DOFs to a 3D virtual environment display space supporting 6 DOFs, the BeyondDeskVR system utilised the ray casting metaphor (Lu, Yu, and Shi 2020). The finger's world coordinate (X_W, Y_W) on the physical desktop was transferred to the cursor position (X_V, Y_V, $Depth$) on a virtual plane located in front of the user at a distance of $Depth$ from the user's eye centre at (0, 0, 0). The user's spatial pointing direction (X_V, Y_V, $Depth$) was defined as the vector that originates from the user's eye centre at (0, 0, 0) and terminates at the virtual cursor (X_V, Y_V, $Depth$). The user's pointing target was determined by the direction of the ray. To distinguish between pointing gestures and other operational gestures, we activated the cursor pointing state when the user places a single finger on the desktop. Users could adjust the ray casting angle by sliding their fingers to point toward the target. Target selection and control could be completed by performing specific finger gestures.

However, achieving high precision through ray pointing alone was challenging in practical operations. We sought inspiration from the interaction logic of touchscreens in the real world (Johnson 1965) and optimised the pointing method accordingly. Due to the inherent ambiguity of fingertip touches, explicit touchscreen interactions in the real world often determine pointing targets on the basis of the overlap between the touch area and screen control elements. To improve the precision of the ray-casting method, we utilised a volume-based pointing technique (LaViola et al. 2017). Specifically, we opted for sphere-based

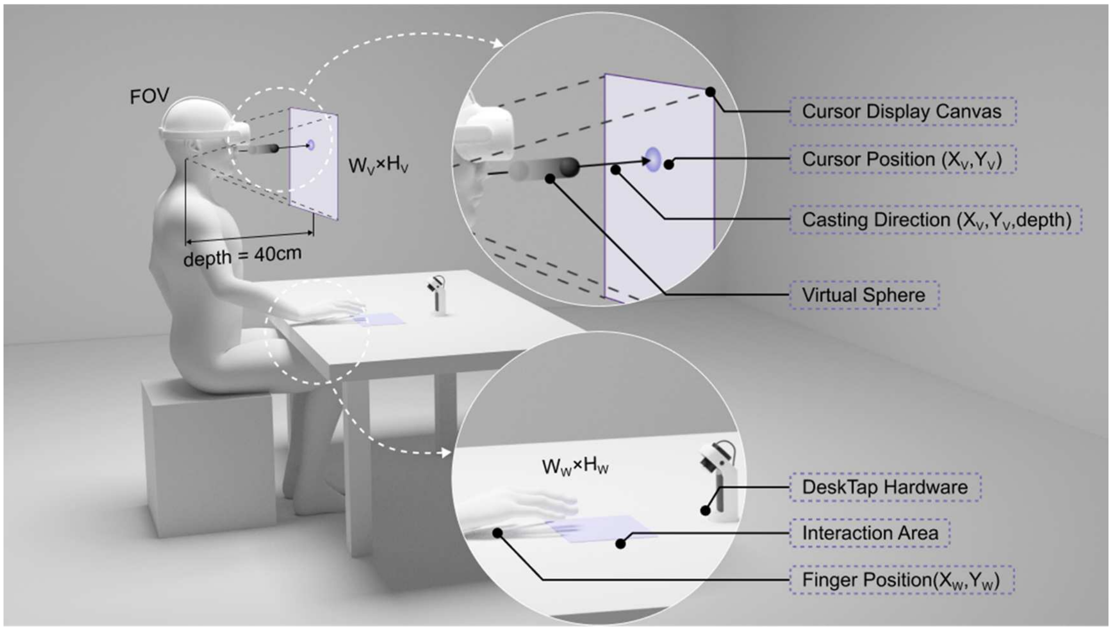


Figure 4. Diagram of control-display mapping.

ballistic casting over ray casting by projecting an invisible sphere of a predefined size on the basis of the user's pointing direction. The user's pointing target was defined by checking the collision or overlap status between the sphere and the target.

Finally, the control-display mapping pointing method based on the sphere-casting metaphor is shown in Figure 4. A virtual canvas was situated 40 cm away from the user, and its dimensions were determined by the user's eye-camera setup's field of view (FOV). The virtual cursor was located within this virtual canvas and the interaction area was situated within the comfort interaction region as determined in the preliminary stages. Unlike the displacement-based mapping used for touchpads or screen cursors on personal computers, BeyondDeskVR used a proportional mapping relationship between the finger position (X_W, Y_W) and the virtual cursor position (X_V, Y_V), as expressed in Equation (5). Here, W_W and H_W represent the width and height, respectively, of the rectangular area for finger movement on the desktop, whereas W_V and H_V represent the width and height, respectively, of the rectangular area on the virtual cursor displayed on the display canvas.

$$\begin{bmatrix} X_V \\ Y_V \end{bmatrix} = \begin{bmatrix} \frac{W_V}{W_W} & 0 \\ 0 & \frac{H_V}{H_W} \end{bmatrix} \begin{bmatrix} X_W \\ Y_W \end{bmatrix} \quad (5)$$

When the user's finger touched the desktop, an invisible virtual sphere with a diameter of 2 cm was released from the centre of the user's eye into the virtual space and directed towards the user's pointing direction $(X_V, Y_V, Depth)$. The interaction target was considered correctly pointed to once the virtual sphere collided with the target during its casting.

3.5. Definition of selection

The definition of selection aims to design appropriate gestures to trigger the target selection process after pointing to the target. In the BeyondDeskVR system, we focused on defining methods for selecting desktop virtual hand interactions for basic target selection tasks. We proposed three feasible selection gestures for pointing-selection tasks: one-finger tap, one-finger dwell, and two-finger tap. The interaction logic for these three selection methods is shown in Figure 5.

The one-finger tap, as shown in Figure 5(a), followed an interaction logic where the user completes target pointing and then selects the target by lifting and clicking the single finger again.

The one-finger dwell, as shown in Figure 5(b), followed an interaction logic similar to that of gaze-based input (Blattgerste, Renner, and Pfeiffer 2018) in the form of dwell triggering. In this approach, after pointing to the target with a single finger, the user maintains their finger in its current position for a continuous 0.5 s to finalise the target selection.

The two-finger tap, as shown in Figure 5(c), followed an interaction logic where the user points to the target with one finger and then presses the second finger to select the target.

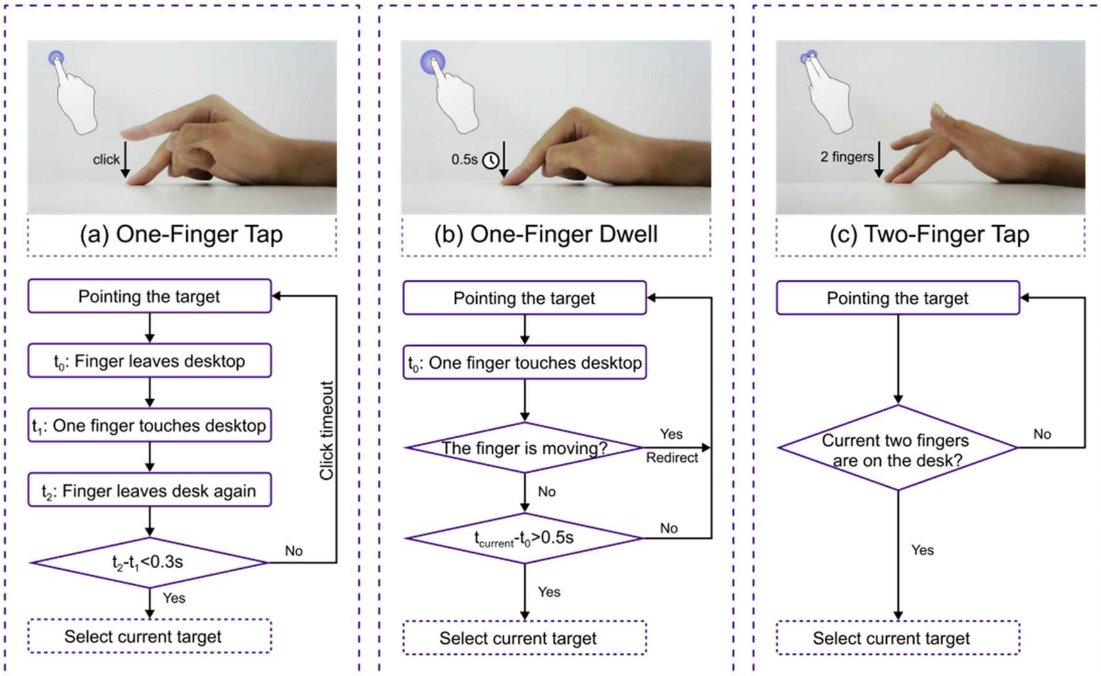


Figure 5. The interaction logic for three desktop gesture selection methods.

3.6. Extended virtual hand interaction methods

The development of the extended virtual hand interaction methods aims to construct an extended virtual hand interaction prototype from two perspectives – UI interaction tasks and object manipulation tasks – to explore seamless switching and practical applications involving mid-air hand interactions and desktop gesture interactions in common interaction scenarios. In this section, we focus on the design of functionalities related to the most common UI interaction tasks and 3D object manipulation tasks. By addressing interaction targets such as 3D user interfaces and virtual objects and tasks such as pointing selection and object manipulation, we investigated the application potential and suitability of the BeyondDeskVR system in real desktop scenarios. The entire application prototype[1] for the extended virtual hand interaction was built on Unity 2021.3.6f1c1 utilising the Oculus Integration plugin for mid-air hand interaction and the methods described in the previous sections for desktop gesture interactions.

In the application prototype of the extended virtual hand system, common UI interaction tasks included button selection, toggle selection, scrollbar dragging, scroll view dragging, and dropdown menu selection, and common object manipulation tasks included object selection, switching manipulation mode, 6-DOF object translation, 6-DOF object rotation, and object zooming.

For the gesture design of mid-air hand interactions: In UI interaction tasks, referring to the real-world characteristic of direct hand contact with targets, we chose mid-air hand forefinger clicking, dragging UI controls, and other actions to achieve the basic UI interaction function. In object manipulation tasks, inspired by the 3D object manipulation gesture design proposed by Alkemade, Verbeek, and Lukosch (2017), we chose gestures such as finger pinching and both hands pinching, which emulate natural physical interactions, to achieve object translation, rotation, and zooming.

For the gesture design of desktop gesture interactions: In UI interaction tasks, referring to the gestures used for virtual cursor control on physical touchpads, we achieved pointing, selecting, and dragging of UI controls by single-finger movement, hovering, and the selection gestures designed in the previous section. In object manipulation tasks, due to the multiple DOFs and types of manipulation modes involved, it was challenging to accomplish complete 3D object manipulation through a single finger. We referred to the multitouch gesture design for the high-DOF manipulation of 3D objects with handheld touch devices (Goh, Sunar, and Ismail 2019). Adhering to the principle that the translation, rotation, and zoom modes did not conflict with each other, we strived to achieve object translation, rotation, and zooming through multitouch gestures involving three or fewer fingers (Brouet, Blanch, and Cani 2013).

The detailed descriptions of the implementation methods for the mid-air hand interaction and desktop gesture interaction for each interaction function are

Figure 6. UI interaction and object manipulation prototypes of BeyondDeskVR.

shown in Figure 6 and Table 1. It is important to note that the 'Manipulation Mode Switch' refers to the mode transition between three object manipulation states: translation, rotation, and zooming.

In summary, by constructing desktop gesture recognition hardware, recognising desktop finger touch states, acquiring the world coordinates of the touching fingers, designing pointing and selection interaction strategies, and establishing extended virtual hand interaction methods, we explored the combined applications of desktop gesture interactions and mid-air hand interactions of BeyondDeskVR in various interaction tasks, such as pointing and selecting, 3D UI interaction, and 3D object manipulation.

4. Experiments

In this section, we used the most fundamental pointing and selection task to comprehensively evaluate the performance of desktop gesture interaction and mid-air hand interaction in BeyondDeskVR, assessing them in terms of subjective assessments, task performance, and interaction fatigue. This study conducted validation experiments based on the Fitts click paradigm of the

Table 1. The description of extended virtual hand interaction methods.

Interaction types	Interaction tasks	Mid-air hand interaction	Desktop gesture interaction
UI Interaction	Button Selection	Index finger touch	Two-finger tap
	Toggle Selection	Index finger touch	Two-finger tap
	Scrollbar Drag	Index finger drag	One-finger dwell
	Scroll View Drag	Index finger drag and touch	One-finger dwell, and two-finger tap
	Dropdown Selection	Index finger touch	Two-finger tap
Object Manipulation	Object Selection	Pinch gesture for selection	One-finger dwell
	Manipulation Mode Switch	No	Three fingers transformation to four fingers
	Object Translation	Pinch gesture and palm movement	Two-finger dwell, and three-finger dwell
	Object Rotation	Pinch gesture and wrist rotation	Two-finger rotation
	Object Zoom	Two-hand pinch, expand or retract	Two-finger spread and pinch
		Mid-air hand pitch for selection, and the two-finger spread and pinch on the desktop for zoom.	

ISO 9241-411 standard (Iso 2012) to assess the interaction performance of three desktop gesture selection methods (one-finger tap, one-finger dwell, two-finger tap) and the mid-air hand selection method (virtual hand touch). Additionally, we evaluated the user experience and fatigue associated with all four selection methods through the SUS (Brooke 2013), the NASA-TLX (Hart and Staveland 1988), a subjective preference ranking questionnaire, and consumption endurance.

4.1. Participants and apparatus

The experiment involved a total of 36 participants, consisting of 20 males and 16 females aged 22–25 years, recruited from Southeast University. All the participants had normal corrected vision and were right-handed. More than half of them had prior experience with VR before the experiment. Before the experiment began, all the participants received sufficient training to familiarise themselves with the VR environment. The study was approved by the IEC for Clinical Research of Zhongda Hospital, Affiliated with Southeast University.

The experiment was conducted in a monochromatic VR environment without depth cues to minimise interference with user interaction. The experimental scene was built via Unity 2021.3.6f1c1, and the data were collected via C# scripts. An Oculus Quest 2 head-mounted display was used as the VR display device. The mid-air hand interaction uses the cameras on the Oculus headset to capture hand gestures, with recognition performed through the Oculus Integration plugin. Desktop gesture interaction is implemented via the desktop gesture recognition device, as described in the technical section of Chapter 3. The experimental procedures were conducted on a computer equipped with an AMD R7-5800X CPU and an RTX 3080 GPU.

4.2. Procedure

Before the start of the experiment, the experimenter provided participants with a detailed introduction to the experimental procedures and the clicking tasks. The participants put on the Oculus head-mounted display and became familiarised with the virtual environment and the specific clicking tasks related to the subsequent evaluation. Each participant had to practice every interaction method for at least nine trials, confirming their full proficiency and readiness verbally before moving on to the formal experimental stage.

The experimental environment was developed based on the Fitts click paradigm (MacKenzie 1992) of the ISO 9241-411 standard. It consisted of a circular arrangement comprising nine grey circular targets, with the highlighted red target to be clicked. The target depth was set at 40 cm (Katzakis et al. 2021; Lubos et al. 2016). Considering the comfortable interaction region for human eyes and hands (Penumudi et al. 2020), the centre of the circular arrangement was placed 5° below the centre of the VR straight-ahead field of view. The experimental setup is shown in Figure 7. Notably, the above targets are fixed in front of the user's head and move with the user's head.

The experimental task for a single trial was as follows. First, participants were prompted to start a new trial by using their index finger or one of three desktop gestures to click the start button located at the centre of the circular layout. Once initiated, the trial's starting time was recorded, and a scene with nine circular targets appeared within the participant's field of vision. The participants were subsequently instructed to click the red-highlighted target as quickly as possible while maintaining accuracy.

During each trial, the click time and the click position were recorded for each target that clicked in the circular layout. Due to the distinctions between mid-air hand interactions and desktop gesture interactions, there were discrepancies in the methods utilised to record click positions. For mid-air hand interactions, the click position was defined as the three-dimensional coordinate of the centre of the collision sphere, which

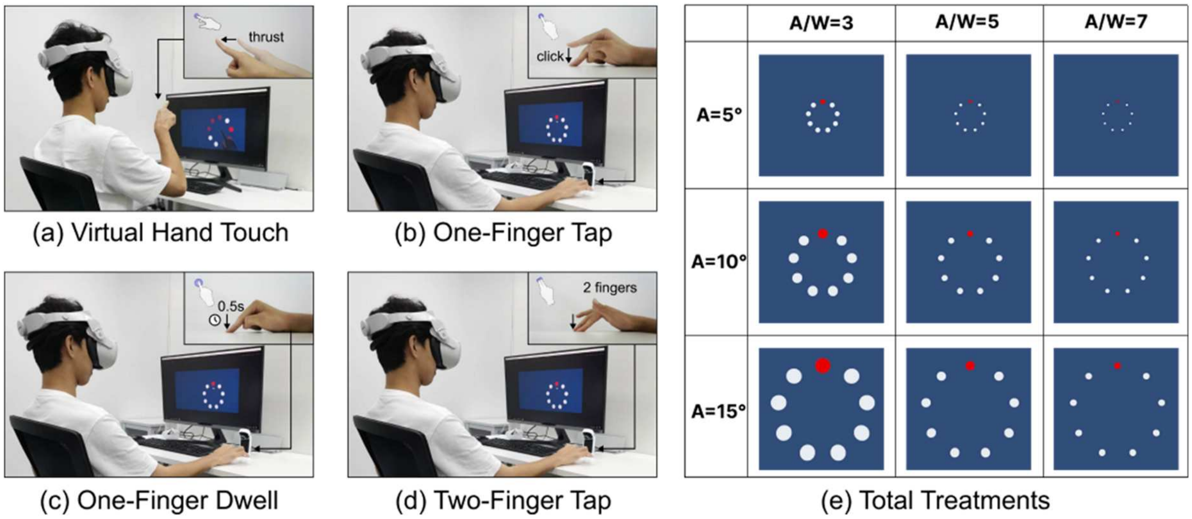


Figure 7. Diagram of the experimental scene.

is attached to the user's index finger at the moment of contact with the target. For desktop gesture interaction, the click position was defined as the three-dimensional coordinate of the collision point between the sphere ray-cast and the circular target.

The participants were required to complete all the clicking tasks via four selection methods (virtual hand touch, one-finger tap, one-finger dwell, two-finger tap). A 4×4 Latin square design was used to plan the execution order of each selection method for each participant to eliminate sequence effects. The experiment for each selection method consisted of a total of 9 treatments, specifically, 3 levels of movement distances ($A = 5^\circ/10^\circ/15^\circ$) and 3 levels of ratios of target size to moving distance ($A/W = 3/5/7$, representing the index of difficulty (ID), which can be calculated via Fitts's law, where $ID_{fitts} = \log_2\left(\frac{A}{W} + 1\right)$ (Soukoreff and MacKenzie 2004). For these nine treatments, a 9×9 Latin square design was also used to eliminate sequence effects between participants, with each treatment repeated three times. After completing the clicking tasks for each selection method, the participants were asked to complete the SUS and NASA-TLX questionnaires to evaluate the user experience of the current selection method. After completing all the tasks, each participant was asked to rank their preferences for the four selection methods. During the formal experiment, the participants completed a total of $4 \times 3 \times 3 \times 3 = 108$ trials, and the entire experiment lasted approximately one hour. Three participants were subsequently selected for repeated measurements of consumed endurance (CE) (Hincapié-Ramos et al. 2014) following the completion of the formal experiment.

4.3. Results

A total of 34,992 click data points were collected in this experiment; 3888 trials were completed by 36 participants, each of whom completed 108 trials. These click data points included both the time and the coordinate data for each individual click. Additionally, each participant provided responses to the SUS and NASA-TLX questionnaires for all four selection methods, along with their subjective preference rankings. Furthermore, arm fatigue for each trial was quantified by consumed endurance (CE).

4.3.1. Subjective questionnaire

The subjective evaluation section collected responses from all 36 participants, each of whom provided feedback on the subjective preference ranking, the SUS questionnaire, and the NASA-TLX questionnaire for the four selection methods. Overall, the user subjective evaluations ranked from best to worst were as follows: one-finger dwell $\geq$ two-finger tap > virtual hand touch > one-finger tap.

The user preferences are shown in Figure 8(a). The rankings of the four interaction methods, as obtained from the users' comprehensive ranking questionnaires, were weighted, with ranks 1–4 corresponding to scores of 4–1, respectively. From the average weighted scores, user preferences are ranked as follows: one-finger dwell = two-finger tap > virtual hand touch > one-finger tap.

The overall SUS and NASA-TLX scores for the four selection methods are shown in Figure 8(b) and (c). One-way ANOVA revealed that there was a significant main effect of the selection methods on the SUS score ($F(3,140) = 32.888$, $p < 0.001$) and overall weighted NASA-TLX score ($F(3,140) = 14.650$, $p < 0.001$). From

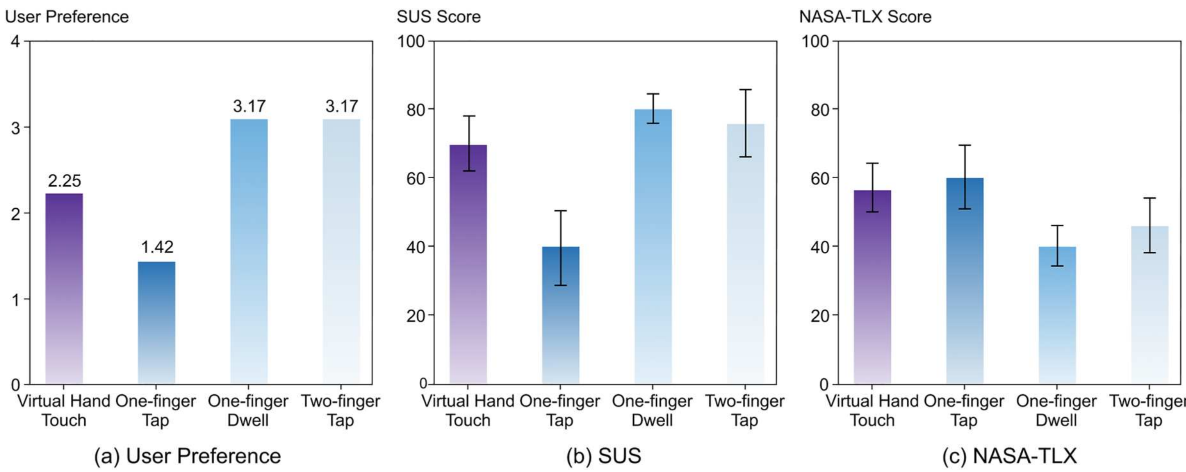


Figure 8. The user preference, SUS and NASA-TLX scores by selection methods.

a gender perspective, there was no significant gender difference in SUS scores (F(1,142) = 1.325, $p = 0.252$), but a significant difference in NASA-TLX scores (F(1,142) = 16.485, $p < 0.001$). The usability score rankings were as follows: one-finger dwell (M = 79.9, SD = 12.7) > two-finger tap (M = 74.0, SD = 20.3) > virtual hand touch (M = 69.5, SD = 17.2)> one-finger tap (M = 39.9, SD = 22.9). The overall fatigue score rankings were as follows: one-finger dwell (M = 39.8, SD = 13.2) < two-finger tap (M = 43.9, SD = 15.0) < virtual hand touch (M = 56.5, SD = 14.1) < one-finger tap (M = 60.2, SD = 18.5).

4.3.2. Task performance

On the basis of the experimental records of 3888 trials, which included the time and position of target clicks, our analysis focused primarily on three aspects: movement time (MT), click position offset (accuracy), and throughput (TP) across different experimental treatments.

The movement times and accuracies of the four selection methods are shown in Figure 9. Movement time was calculated as the temporal discrepancy between the current and the previous target click moments. This is because the distance of the first click was the distance between the centre button and the first target, which was half that of the 2nd to 9th clicks. Therefore, we discarded the first click to ensure the same moving distance. The average time for the 2nd to 9th clicks in each round was recorded as the MT for that trial. One-way ANOVA revealed that there was a significant main effect of the selection methods on MT (F(3,3884) = 922.564, $p < 0.001$). Additionally, for each selection method, the movement time increased with increasing ID. There was no significant difference in MT between the one-finger dwell (M = 1309 ms, SD = 278 ms), two-finger tap (M = 1184 ms, SD = 384 ms), and virtual hand touch conditions (M = 1134 ms, SD = 374 ms), but the one-finger tap (M = 2129 ms, SD = 741 ms) method took more time than the other methods did. Accuracy was calculated as the ratio of the distance between the actual click position and the centre of the click target to the width of the target. One-way ANOVA revealed that there was a significant main effect of the selection methods on click accuracy (F(3,3884) = 922.564, $p < 0.001$). The click offset percentage increased with increasing ID, and the differences in accuracy among the four methods decreased. Overall, the click offset of one-finger tap (M = 59.7%, SD = 10.1%) was significantly greater than that of two-finger tap (M = 54.5%, SD = 11.4%), one-finger dwell (M = 49.2%, SD = 13.8%), and virtual hand touch conditions (M = 44.3%, SD = 13.4%). Additionally, from a gender perspective, there was a significant difference between males and females in MT (F(1,3886) = 11.068, $p < 0.001$). The MT for males (M = 1409 ms, SD = 616 ms) was slightly lower than that for females (M = 1476 ms, SD = 634 ms). However, there was no significant difference in accuracy (F(1,3886) = 3.109, $p = 0.078$).

Fitts' law provides a human movement prediction and measurement model for HCIs and ergonomics. It introduces throughput (TP) as a quantifiable measure of human performance that is independent of movement amplitude (A) and target distance (W). The TP is a comprehensive metric that affects interaction accuracy and speed. ISO 9241-411 provides the most widely adopted method for calculating the TP, which is as follows:

$$TP = \frac{ID_{Shannon}}{MT} = \frac{\log_2\left(\frac{A_e}{W_e}+1\right)}{MT} = \frac{\log_2\left(\frac{A_e}{4.133 \times SD_{dx}}+1\right)}{MT} \quad (6)$$

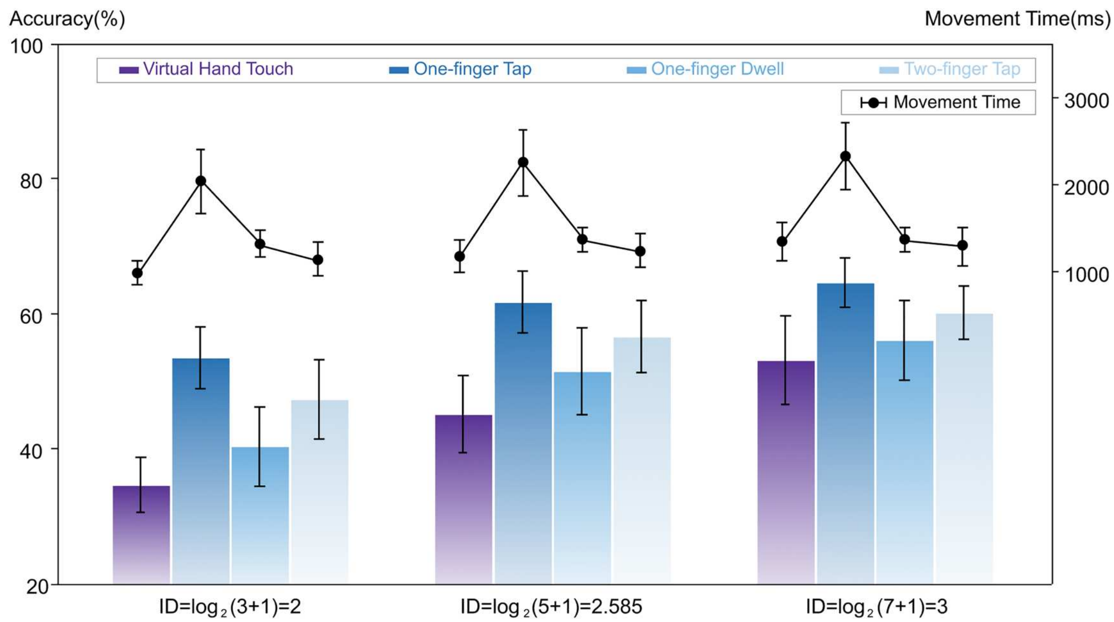


Figure 9. Movement time and click accuracy by selection methods and IDs.

Here, A_e represents the effective target amplitude, which is the average of the user's actual movement distances, quantified by the distance between the two targets; dx represents the projection interpolation of the actual click location to the target location along the task axis; SD_{dx} represents the standard deviation of this interpolation; W_e represents the effective target width, which is equivalent to $4.133 \times SD_{dx}$; and MT represents the average time required to complete the click task. Using the above formula, we calculated the throughputs for mid-air hand interaction, one-finger tap, one-finger dwell, and two-finger tap as 2.80, 1.30, 2.32, and 2.38, respectively.

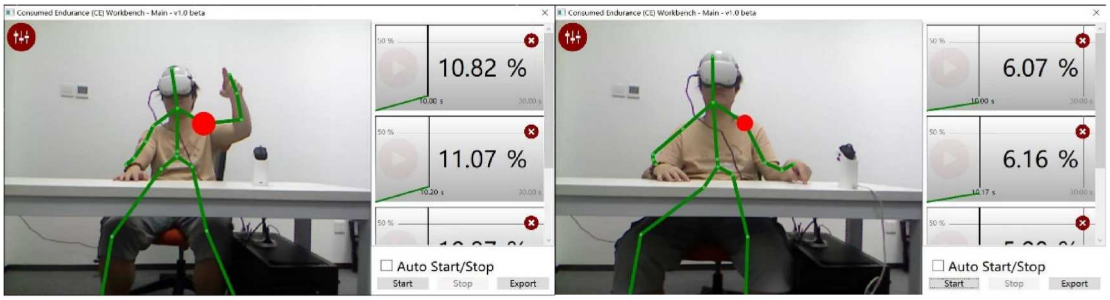
4.3.3. Arm fatigue

Arm fatigue was quantified via consumption endurance (CE) metrics. Consumed endurance (Hincapié-Ramos et al. 2014) is a biomechanical metric used to quantify upper arm fatigue during mid-air gesture interactions on the basis of the biomechanical structure of the human arm. The system tracked the bending and stretching of the arm over time using noninvasive camera-based skeletal tracking systems, thereby calculating the cumulative fatigue of human interaction movements. In this study, the skeletal information of users during both mid-air hand and desktop gesture interactions was captured via a Microsoft Xbox 360 Kinect sensor, and the CE was quantified at 10-second intervals. The results from three participants indicated that the consumed endurance for desktop gestures ($M = 6.04\%$, $SD = 1.16\%$) was significantly lower than that for mid-air hand interactions ($M = 9.82\%$, $SD = 0.56\%$). Figure 10 shows the consumed endurance for 10-second intervals during mid-air hand and desktop gesture interactions for one participant. The average consumed endurance for 10 s during mid-air hand interaction was 10.95%, while during desktop gesture interaction, it was 6.07%. It should be noted that during desktop gesture interaction, the desk provided support for the arm, leading to lower actual consumed endurance than what the test data suggested.

5. Discussion

This paper describes a BeyondDeskVR system designed to complement various types of mid-air hand interactions, addressing the fatigue issues inherent in existing mid-air hand interactions. The BeyondDeskVR system utilised infrared laser projection sensing technology to enable stable desktop gesture inputs. We employed an indirect metaphorical approach to map user gestures in a small desktop space to a 3D virtual cursor, enabling seamless integration with mid-air hand interactions for desktop virtual reality interactions. Ergonomic evaluations of both the desktop gesture interaction of the BeyondDeskVR system and mid-air hand interaction were conducted, with the NASA-TLX workload and CE indicators demonstrating that the BeyondDeskVR system significantly reduces fatigue compared with mid-air interactions, effectively alleviating user fatigue during interaction processes.

The results showed that all 36 participants were able to proficiently master the use of the BeyondDeskVR system after a relatively short learning period, indicating



(a) (b)

Figure 10. CEs for 10-second intervals by virtual hand and desktop gesture interactions. (a) CE during virtual hand interactions. (b) CE during desktop gesture interactions.

that the system is easy to learn and has a certain level of user acceptance. The experiment compared the performance differences of three selection gestures in desktop gesture interactions (one-finger tap, one-finger dwell, and two-finger tap) with those in mid-air hand interactions, with a focus on the pointing and selection tasks. The evaluation was conducted from subjectivity, performance, and fatigue aspects to quantify the effects.

The subjective assessment results of the experiment indicate that user preferences, SUS scores, and NASA-TLX scores exhibit consistency. One-finger tap was not included in the subsequent analysis due to the high error rate caused by the Heisenberg effect (Wolf et al. 2020) during the tapping process. One-finger dwell and two-finger tap in desktop gesture interactions outperformed mid-air hand interactions. This is mainly because the majority of participants reported that arm fatigue associated with mid-air interactions was intolerable during an extended period. The post session interviews indicated that significant fatigue effects typically emerged after approximately three minutes of engaging in airborne gesture interactions. These findings suggested that in high-frequency and prolonged operation tasks, user fatigue and comfort are crucial factors influencing the user experience. Despite the natural advantage of consistency between control and display in mid-air hand interactions, the fatigue caused by static muscle loads prompts users to explore alternative interaction methods.

The task performance of the experiment was assessed on the basis of two metrics, accuracy and movement time, both of which exhibited strong collinearity. The results indicated that there were no significant differences in performance between one-finger dwell, two-finger tap and virtual hand touch, whereas the performance of the one-finger tap was inferior. This phenomenon is closely associated with the Heisenberg effect (Wolf et al. 2020) during the selection process. The selection process typically involves two sequential subtasks: pointing and selecting. The Heisenberg effect reflects the deviation in the direction of target pointing caused by the motion of the selection action (Bowman et al. 2001) during limb-based interactions, which ultimately results in greater selection offset (Wolf et al. 2020), greater error rates (Weller et al. 2021), or longer selection times (Brasier et al. 2020). One-finger dwell and two-finger tap motions effectively separate pointing and confirming in two stages via time thresholds and gesture spatial states, thus improving task performance and mitigating the Heisenberg effect. This result suggested that when designing interaction gestures triggered by a limb's free movement, the effective separation of pointing and confirming should be considered. Additionally, male participants exhibited shorter movement time compared to female participants, reflecting a greater emphasis on efficiency during interaction.

The throughput of one-finger dwell and two-finger tap in desktop interactions was comparable to that during mid-air interactions. Throughput can be considered a comprehensive metric reflecting the accuracy and performance of an interactive system, and it is independent of specific user operations (Zhai 2004). Compared with the TPs of other interaction methods, the TP of the BeyondDeskVR system is comparable to those of three-dimensional selection methods, such as the ray metaphor (TP = 2.6) (Mifsud et al. 2022), ViewfinderVR (TP = 1.6~3.2) (Kim and Xiong 2022) and Touching the Sphere (TP = 2.0) (Lubos et al. 2016), despite being surpassed by two-dimensional selection methods such as the mouse (TP = 3.7) and touchpad (TP = 6.9) methods (Scott Mackenzie 2015). The similarity in throughput between the BeyondDeskVR system and various mid-air hand interactions suggests that they can be used to address interfaces requiring similar precision, within the same interaction system.

As the upper limbs account for only 6% of the total body weight (Plagenhoef, Evans, and Abdelnour 1983), repetitive and sustained lifting of the arms during user interactions in mid-air can easily lead to prolonged muscle contractions and restricted blood flow. With a decrease in oxygen levels in the blood, muscle cells switch their energy source from aerobic metabolism to glycolysis. However, due to the limited glucose storage in muscles, cells can only generate energy for a short period to maintain contraction. As soon as energy is exhausted, users experience significant fatigue (Hincapié-Ramos, Guo, and Irani 2014). While interacting with the BeyondDeskVR system, users take advantage of the physical support of the desktop device to avoid prolonged muscle contractions in their arms. The reduced interaction area converts larger shoulder and arm movements into smaller finger and wrist motions, mitigating the extended stress on major muscle groups. This indirectly highlights the importance of aligning input positions with biomechanical ergonomic principles to achieve an improved user experience in VR (Penumudi et al. 2020).

In summary, the four selection methods evaluated in the BeyondDeskVR system each have their own characteristics. One-finger tap needs to be excluded initially due to the lack of separation between pointing and confirmation actions, resulting in the Heisenberg effect. One-finger dwell, in the temporal dimension, distinguishes pointing from selection on the basis of the stationary time of the finger. Users can complete target selection by simply keeping their fingers stationary on the desktop without requiring additional movements. Therefore, it possesses lower gesture complexity and higher learnability, albeit with a slightly longer overall interaction cycle. Two-finger tap, in the spatial dimension, distinguishes pointing from selection by the number of pressed fingers, and the interaction is more rapid. Virtual hand touch offers the highest DOFs of gesture interaction, but arm fatigue during prolonged interactions should not be overlooked.

The BeyondDeskVR system represents a contact-based gesture interaction, while mid-air hand interaction represents a noncontact gesture interaction, demonstrating their natural complementary advantages. Noncontact interaction, represented by mid-air hand interaction, employs a direct metaphor interaction method. While it offers advantages such as high DOFs, naturalness, intuitiveness, and immersion, it also presents issues such as arm fatigue during prolonged interactions and reduced user comfort. The BeyondDeskVR system employs an indirect metaphor by utilising physical surface support to reduce fatigue, increase interaction speed, and enhance comfort. However, this inevitably leads to a lack of intuitive perception during the interaction process. Therefore, we combined the advantages of desktop interaction and mid-air interaction and demonstrated that complementary benefits can be achieved while preserving the benefits of bare-hand interaction.

With respect to the prototype application of BeyondDeskVR, establishing compatibility in terms of control and display is crucial for the usability of indirect metaphors. To establish mappings between the desktop 2D control space and the VR 3D display space, the primary consideration for the BeyondDeskVR system in integrating multimodal interactions within a single system is ensuring the naturalness and smoothness of modal switching, as well as providing consistent support for interactions with the same user interface. Both mid-air and desktop virtual hand interactions utilise gestures to interact with 3D interfaces or objects. Therefore, their motion designs should maintain consistency with the forms of movement to facilitate faster user learning. Additionally, different movement quantities exist due to differences in the input space (Zhou and Bai 2023). With the larger input space for mid-air gestures, the range of motion for mid-air interactions can be designed to be larger, completing tasks through the entire hand or dual-hand actions. In contrast, desktop gestures are designed to complete all interaction operations with only minimal finger movements to improve comfort. The comparable throughput between desktop gestures and mid-air hand interactions supports their application within the same precision human–computer interaction interface and the same cursor feedback system. From the perspective of system input, the desktop interaction in this system limits the input space within the small, comfortable interaction range of the human hand. It was designed to utilise the two or three most flexible fingers for interaction. From the perspective of control and display mapping, this system addresses nonuniform display and control space sizes by proportionally mapping the finger position and spatial 3D cursor pointing. Additionally, the inconsistent display and control dimensions are resolved by semantically mapping multiple touch gestures and interaction feedback.

The BeyondDeskVR system introduced a novel interaction paradigm to improve the user experience in VR, serving as an extension of existing mid-air virtual hand interactions. This system combines the convenience of bare-hand interaction with the low fatigue advantage of desktop interaction devices like mice. Moreover, the consistent bare-hand interaction and similar operational precision prevent a sense of disconnection when switching between mid-air hand interactions and the BeyondDeskVR system. Compared with other contact-based desktop interaction devices, BeyondDeskVR physically separates devices from the user's hands, enabling cost-effective extraction of desktop gestures while maintaining bare-hand input. Compared with single mid-air hand interactions, integrating the BeyondDeskVR system expands the 3D interaction space to the desktop, effectively reducing user fatigue by transferring and narrowing the input space, while constantly providing tactile feedback during interaction.

From the design perspective of the interaction system in this study, the 3D virtual environments represented by VR present the typical characteristics of high-input DOFs and high-output DOFs. Therefore, 3D interaction systems should not be confined to a single interaction form or control-display mapping relationship. BeyondDeskVR not only proposes a new interaction device but also introduces a new interaction method by combining 3D mid-air gestures with 2D contact-based gestures on any supporting surface. This approach meets the dual needs of naturalness and comfort in VR. Regarding the input, the high DOFs in the control space imply the possibility of choosing diverse input devices and forms. Regarding the output, the high DOFs in the display space imply the potential for presenting more 3D user interfaces and intuitive feedback effects. Therefore, combining direct and indirect metaphors and mapping high DOFs in input and output forms, along with their fusion with interaction contexts and tasks, serve as a valuable approach to enhancing the user experience in 3D interaction systems.

6. Limitations and future work

Despite achieving commendable performance in basic pointing and selection tasks, the BeyondDeskVR system has room for optimisation in several areas. First, our system adopted a visual sensing-based solution and was susceptible to issues such as occlusion and environmental lighting effects. In this study, we utilised the Otsu algorithm (Otsu 1979) to reduce the impact of ambient light, and users were advised to use this system in indoor settings without direct sunlight. Second, the BeyondDeskVR system was designed for use on any flat surface; however, infrared light paths could be obstructed if objects were present. Finally, the mid-air gesture sensor and desktop gesture sensor of our system are two separate and independent devices. In the future, integrating these devices into the camera of the head-mounted display would minimise hardware limitations and simultaneously address the aforementioned occlusion issues. Furthermore, future research will explore the system's usability across a broader age range beyond youth users.

7. Conclusion

To address the fatigue associated with gesture interaction in a seated desktop context, this paper introduces BeyondDeskVR, a desktop gesture interaction system designed for seamless integration with existing virtual hand interaction systems. We extended the input space of virtual hands to any desktop, offering low fatigue, adequate precision, and speed, making it suitable for prolonged use in seated environments without the need for wearable devices. BeyondDeskVR utilised infrared laser projection sensing principles to construct a desktop gesture recognition device and utilised image processing algorithms to effectively recognise the touch status on the desktop. Our system developed an indirect metaphor-based control and display mapping method for desktop interaction, mapping user desktop gestures to remote 3D cursor pointing and system control, enabling its application in typical tasks such as 3D UI interaction and 3D object manipulation. Ergonomic evaluations indicated a user preference for the BeyondDeskVR system, with significantly lower fatigue levels than those associated with mid-air hand interaction. Its comparable operational precision allows it to be effectively combined with various existing virtual hand interaction methods. BeyondDeskVR serves as an effective supplement to current virtual hand interactions in seated desktop scenarios, offering a new approach for expanding virtual reality applications.

Notes

1. The demonstration video of the application prototype can be accessed at https://drive.google.com/file/d/15m0nlETw7DChDeTUgcwCsFU1kUZ_uF2U/view?usp=sharing

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Appendix

Table A1. The average movement time by different treatments (ms).

Method		Virtual Hand	One-finger Tap	One-finger Dwell	Two-finger Tap
Average		1134	2129	1309	1184
A = 5°	A/W = 7	1395	2473	1390	1319
	A/W = 5	1260	2456	1410	1295
	A/W = 3	948	2338	1482	1189
A = 10°	A/W = 7	1268	2129	1250	1180
	A/W = 5	1071	2085	1234	1132
	A/W = 3	950	1846	1156	1072
A = 15°	A/W = 7	1237	2134	1336	1222
	A/W = 5	1082	1978	1310	1181
	A/W = 3	998	1720	1210	1067

Table A2. The average offset percentage by different treatments (%).

	Method	Virtual hand	One-finger tap	One-finger dwell	Two-finger tap
	Average	44.28	59.54	49.17	54.51
A = 5°	A/W = 7	67.76	67.81	65.15	64.26
	A/W = 5	57.49	69.48	65.87	66.21
	A/W = 3	41.49	63.83	53.60	59.53
A = 10°	A/W = 7	49.00	63.89	52.43	58.53
	A/W = 5	41.34	61.97	47.54	56.14
	A/W = 3	33.99	50.43	36.08	43.06
A = 15°	A/W = 7	42.63	61.52	50.35	57.36
	A/W = 5	36.35	53.29	40.48	46.83
	A/W = 3	28.82	45.06	31.26	38.90

Table A3. The average throughput by different treatments.

Method	A (°)	A/W	A (cm)	W_e (cm)	ID_{fitts} (bits)	$ID_{Shannon}$ (bits)	MT (ms)	TP (bits/s)
Virtual Hand Touch	5	3	6.867	1.334	2.00	2.62	948	2.76
	5	5	6.867	1.087	2.58	2.87	1260	2.28
	5	7	6.867	0.913	3.00	3.09	1395	2.22
	10	3	13.680	2.236	2.00	2.83	950	2.98
	10	5	13.680	1.606	2.58	3.25	1071	3.04
	10	7	13.680	1.314	3.00	3.51	1268	2.77
	15	3	20.390	2.843	2.00	3.03	998	3.04
	15	5	20.390	2.090	2.58	3.43	1082	3.17
	15	7	20.390	1.668	3.00	3.73	1237	3.01
Average								2.80
One-finger Tap	5	3	6.867	2.169	2.00	2.06	2338	0.88
	5	5	6.867	1.381	2.58	2.58	2456	1.05
	5	7	6.867	0.945	3.00	3.05	2473	1.23
	10	3	13.680	3.223	2.00	2.39	1846	1.30
	10	5	13.680	2.404	2.58	2.74	2085	1.32
	10	7	13.680	1.703	3.00	3.18	2129	1.49
	15	3	20.390	4.461	2.00	2.48	1720	1.44
	15	5	20.390	2.963	2.58	2.98	1978	1.51
	15	7	20.390	2.475	3.00	3.21	2134	1.50
Average								1.30
One-finger Dwell	5	3	6.867	1.746	2.00	2.30	1482	1.55
	5	5	6.867	1.229	2.58	2.72	1411	1.93
	5	7	6.867	0.797	3.00	3.27	1390	2.35
	10	3	13.680	2.333	2.00	2.78	1156	2.40
	10	5	13.680	1.878	2.58	3.05	1234	2.47
	10	7	13.680	1.449	3.00	3.38	1251	2.71
	15	3	20.390	2.990	2.00	2.97	1210	2.45
	15	5	20.390	2.324	2.58	3.29	1310	2.51
	15	7	20.390	2.138	3.00	3.40	1336	2.54
Average								2.32
Two-finger Tap	5	3	6.867	2.034	2.00	2.13	1189	1.79
	5	5	6.867	1.299	2.58	2.65	1295	2.05
	5	7	6.867	0.891	3.00	3.12	1319	2.37
	10	3	13.680	2.946	2.00	2.50	1072	2.33
	10	5	13.680	2.317	2.58	2.79	1133	2.46
	10	7	13.680	1.684	3.00	3.19	1180	2.70
	15	3	20.390	3.900	2.00	2.64	1067	2.47
	15	5	20.390	2.713	2.58	3.09	1181	2.62
	15	7	20.390	2.541	3.00	3.17	1222	2.60
Average								2.38